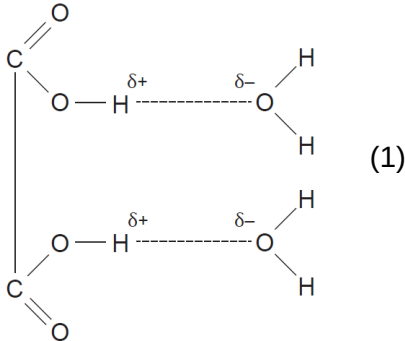


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		$2 \text{ C}_4\text{H}_9\text{OH} + 2 \text{ CO} + \frac{1}{2} \text{ O}_2$		1		1		
		(ii)		award (1) for any of following cost of the catalyst separation of liquid materials from the catalyst cost of (maintaining) very high pressure accept any other valid suggestion		1		1		
		(iii)		ethanedioic acid is able to hydrogen bond to water molecules (hence its solubility) (1)  hydrogen bonding to just one water molecule is acceptable the ester does not contain an O—H bond, so cannot hydrogen bond with water (hence largely insoluble) (1)	1	1				
					1			3		